

b. Suppose that host A is connected to a router R 1, R 1 is connected to another router, R 2, and R 2 is connected to host B. Suppose that a TCP message that contains 900 bytes of data and 20 bytes of TCP header is passed to the IP code at host A for delivery to B. Show the Total length, Identification, Flags, and Fragment offset fields of the IP header in each packet transmitted over the three links. Assume that link A-R1 can support a maximum frame size of 1024 bytes including a 14-byte frame header, link R1-R2 can support a maximum frame size of 512 bytes, including an 8-byte frame header, and link R2-B can support a maximum frame size of 512 bytes including a 12-byte frame header.

Ans:

TCP Packet = 900 (data) + 20 (IP header) = 920 Bytes.

IP Packet = 920 (data) + 20 (IP header) = 940 Bytes

Link A-R1: Can support 1024 bytes including 14-byte frame header (e.g., 1010-byte IP data packet) thus no fragmentation is needed.

Length = 940, ID = x, DF = 0, MF = 0, Offset = 0.

Link R1-R2: Can support 512 bytes including 8-byte header (e.g., 512-8=504 bytes of IP data packet and 504 – 20 (IP header) = 484). Since IP data in a frame must be multiple of 8 bytes:

$8 * x \leq 484 \Rightarrow x \leq 60.5$ bytes. Use $x = 60$ bytes

IP data 1 = $60 * 8 = 480$ Bytes

IP packet size 1 = 480 + 20 = 500 Bytes.

IP data 2 = 920 – 480 = 440 Bytes

IP packet size 2 = 440 + 20 = 460 Bytes.

[1] Length = 500, ID = x, DF = 0, MF = 1, Offset = 0.

[2] Length = 460, ID = x, DF = 0, MF = 0, Offset = 60.

Link R2-B: Can support 512 bytes including 12-byte header (e.g., 512-12=500 bytes of IP data packet and 500 – 20 (IP header) = 480). This implies that no additional fragmentation is needed. Thus

[1] Length = 500, ID = x, DF = 0, MF = 1, Offset = 0.

[2] Length = 460, ID = x, DF = 0, MF = 0, Offset = 60.